📘 **AI/ML Internship – Day 20 Notes & Tasks**

🟡 **Module 4: NumPy (Indexing & Slicing – Deep Dive)**

📌 **Part 1: Why Indexing is Important?**

👉 In real AI/ML:

- You don't use full data
- You select specific parts

👉 Example:

- Select rows
- Filter features
- Extract image regions

📌 **Part 2: 1D Array Indexing**

🔹 **Example**

```
import numpy as np

arr = np.array([10, 20, 30, 40])

print(arr[0])   # First element
print(arr[-1])  # Last element
```

🔹 **Slicing**

```
print(arr[1:3])   # [20, 30]
print(arr[:2])    # [10, 20]
print(arr[2:])    # [30, 40]
```

📌 **Part 3: 2D Array Indexing**

🔹 **Example**

```
arr = np.array([
    [1, 2, 3],
```

```
    [4, 5, 6]
])
```

🔹 **Access Elements**

```
print(arr[0, 1])   # 2
print(arr[1, 2])   # 6
```

🔹 **Access Rows**

```
print(arr[0])   # First row
print(arr[1])   # Second row
```

🔹 **Access Columns**

```
print(arr[:, 1])   # Second column
```

📌 **Part 4: 3D Array Indexing (Important)**

🔹 **Example**

```
arr = np.array([
    [[1, 2, 3], [4, 5, 6]],
    [[7, 8, 9], [10, 11, 12]]
])
```

🔹 **Access Layer**

```
print(arr[0])   # First 2D array
```

🔹 **Access Row inside Layer**

```
print(arr[0][1])   # [4, 5, 6]
```

🔹 **Access Single Element**

```
print(arr[1][0][2])   # 9
```

🔹 **Shortcut Syntax**

```
print(arr[1, 0, 2])   # Same result
```

📌 **Part 5: Advanced Slicing**

🔹 **Step Slicing**

```
arr = np.array([10, 20, 30, 40, 50])

print(arr[::2])   # [10, 30, 50]
```

🔹 **2D Slicing**

```
arr = np.array([
    [1, 2, 3],
    [4, 5, 6]
])

print(arr[:, 1:3])
```

📌 **Part 6: Boolean Indexing (Very Important)**

```
arr = np.array([10, 20, 30, 40])

print(arr[arr > 20])
```

👉 Output:

```
[30 40]
```

🔹 **Multiple Conditions**

```
print(arr[(arr > 10) & (arr < 40)])
```

📌 **Part 7: Real-World Thinking**

👉 In AI:

- Select features → slicing
- Filter data → boolean indexing
- Extract image part → slicing

👉 Example:

- Crop image
- Select dataset rows

🎯 **Tasks for Day 20**

🧠 **Task 1: Theory**

1. What is indexing?
2. What is slicing?
3. Difference between indexing and slicing
4. What is boolean indexing?
5. Why indexing is important in ML?

💻 **Task 2: 1D Practice**

1. Access first element
2. Access last element
3. Slice array

💻 **Task 3: 2D Practice**

1. Access element
2. Print row
3. Print column

💻 **Task 4: 3D Practice**

1. Access layer
2. Access row inside layer
3. Access single element

💻 **Task 5: Slicing**

1. Extract sub-array
2. Use step slicing
3. Slice rows and columns

💻 **Task 6: Boolean Indexing**

1. Filter values > 25
2. Filter values between range
3. Replace values conditionally